1 Visual region extraction

Patch–box intersection-area weights
224 × 224 · no CLS token



Input image I

CLIP ViT-B/16 ❄
visual encoder

Patch features
14 × 14 × 768

YOLO detector ❄

Boxes B
(x_1, y_1, x_2, y_2)

Box-weighted
ROI pooling

Region descriptors
N × 768

Linear projection 🔥
768 → 512

V

2 Semantic bank–region alignment

P_n · 150 × 512
A

V_1, T_1

Object fusion 1 🔥
B

Object fusion 2 🔥
B

V_o

V : N × 512

P_v · 150 × 512
A

Relation fusion 🔥
B

V_r

A TRAIN-only bank preparation & aggregation

B One shared fusion detail; separate parameters

3 Image-specific prompt–region alignment

UPT boxes/classes + YOLO boxes/classes

Class-consistent matching · IoU ≥ 0.5
ROI index i / ordered pair (s, o)

Unmatched slots are omitted.

t_o / t_r + ROI links feed the corresponding Slot MLP.

Object adapter · C 🔥
t_o + linked ROI metadata

Relation adapter · C 🔥
t_r + linked endpoint metadata

I → UPT detector ❄
same input image

Object / relation
phrases

CLIP text encoder ❄
t_o / t_r

Named ports → C

V_o'

V_r'

4 Caption generation

Parallel decoder · D 🔥
V_o' and V_r' remain separate
3 layers · dimension 512 · heads 8
Prefix → next-token distribution

Word embedding + fixed PE

5 Caption reranking

Beam search · width 5
max 20 new tokens

Five candidate captions

Three-score reranking · E
S = 0.3D + 0.4C + 0.3O

Final caption